

# $L$ -functions for all meromorphic $\mathrm{SL}_2(\mathbb{Z})$ modular forms, via manifestly finite contours

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## Abstract

We give a single finite, regulator-free functional realizing in one object  $L$ -functions quasi-periods and period polynomials of meromorphic modular forms on  $\mathrm{SL}_2(\mathbb{Z})$ , with generic poles throughout the upper half-plane  $\mathbb{H}$  and at the cusp of  $\mathrm{SL}_2(\mathbb{Z})$ . Our construction is manifestly finite, preserves holomorphicity. It is a two-segment contour integral with an explicit Hurwitz-zeta  $T$ -kernel, built from any finite base point  $\tau_0 \in \mathbb{H}$  and manifestly finite. For weakly holomorphic forms our  $L$ -function is independent of  $\tau_0$  and of the path, reproduces the for  $L$ -functions of weak modular forms in [1], which previously needed non-holomorphic regularization to be defined. Further, when specifying to  $s \in \{2, \dots, k-2\}$  in the critical strip give the period polynomial, and reproduces Brown's quasi-periods [2]. For forms with interior poles the value depends on the homotopy classes, defined in terms of pole-crossings. There is a natural choice of trivial homotopy class, which renders our  $L$ -functions canonical when the modular form has a pole in the interior of the fundamental domain, generalizing [4].

## 1 A finite $L$ -integral

**Definition 1.1** ( $S$ - and  $T$ -segments). *Let  $\tau_0 \in \mathbb{H}$  and  $f \in F_k$ , let  $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ ,  $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  denote the generators of  $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ , and let  $f^{-1}(\infty)$  be the locus of all poles of  $f \in \mathbb{H}$  (inclusive of the cusp of  $\mathrm{SL}_2(\mathbb{Z})$ ). The  $S$ - and  $T$ -segments at  $\tau_0$  are smooth paths in  $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ ,*

$$\gamma_{\tau_0}^S : -1/\tau_0 \longrightarrow \tau_0, \quad \gamma_{\tau_0}^T : \tau_0 - 1 \longrightarrow \tau_0, \quad (1)$$

running from  $S^{-1}\tau_0 = -1/\tau_0$  and  $T^{-1}\tau_0 = \tau_0 - 1$  to  $\tau_0$ .

Avoiding poles is automatic for  $f \in M_k^!$ , as long as the base-point  $\tau_0$ , is not at a cusp of  $\mathrm{SL}_2(\mathbb{Z})$ , i.e. there is only the trivial homotopy class. When  $f \in F_k$ , and  $f$  has poles at finite and non-zero imaginary heights, integration contours fall into nontrivial homotopy classes.

**Definition 1.2** (Hurwitz  $T$ -kernel). *Let  $\zeta(s, x) = \sum_{m \geq 0} (m+x)^{-s}$  as a function  $\zeta : \mathbb{C}_s \times (\mathbb{C} \setminus \{0\})_x \rightarrow \mathbb{C}$  denote the Hurwitz zeta function (meromorphic in  $s$ , holomorphic in  $x \in \mathbb{H}$ ). We define Hurwitz  $T$ -kernel as*

$$\tilde{k}_T(\tau, s) := \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(1-(k-s), \tau+1). \quad (2)$$

**Definition 1.3** ( $L$ -integral). *Let  $f \in F_k$ ,  $\tau_0 \in \mathbb{H}$ , and  $s \in \mathbb{C}$ . Further, let  $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$  be the  $S$ - and  $T$ -segments at  $\tau_0$ , as in Definition 1.1, and let  $\tilde{k}_T$  be as defined in Definition 1.2. The  $L$ -integral of  $f$  is*

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := e^{-\pi i s/2} \left[ \int_{\gamma_{\tau_0}^S} f(\tau) \tau^{s-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, s) d\tau \right]. \quad (3)$$

We use this  $L$ -integral to define a generalized  $L$ -function:

**Definition 1.4** (*L*-function). Let  $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  be as defined in 1.3. We define the generalized *L*-function of  $f \in F_k$  to be,

$$L(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T). \quad (4)$$

Having defined the *L*-integral and *L*-function, we prove they are manifestly finite for  $f \in F_k$ :

**Lemma 1.5** (Finiteness). For  $f \in F_k$ , both integrals in Definition 1.3 are finite for every  $s \in \mathbb{C}$ , every  $\tau_0 \in \mathbb{H}$ , and every choice of paths  $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$ .

*Proof.* Both segments are compact and lie in  $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ , so  $f$  is continuous on them;  $\tau^{s-1}$  is continuous on  $\mathbb{H}$ , and  $\tilde{k}_T(\tau, s)$  is continuous there since  $\zeta(\nu, \tau + 1)$  is holomorphic in  $\tau \in \mathbb{H}$  (Definition 1.2). A continuous integrand on a compact path is integrable, for every  $s \in \mathbb{C}$ .  $\square$

We now turn to path-dependence and  $\tau_0$ -independence of the *L*-integral and the *L*-function of  $f \in F_k$ . This holds for all  $f \in M_k^!$ :

**Proposition 1.6** (Basepoint and path independence). For  $f \in M_k^!$  the value  $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  is independent of  $\tau_0 \in \mathbb{H}$  and of the paths: with no interior poles,  $\mathbb{H}$  is simply connected, so all admissible contours are homotopic. We write  $L^*(f, s)$  for the common value and  $L(f, s) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s)$ .

*Proof.* With no interior poles the integrands  $f(\tau)\tau^{s-1}$  and  $f(\tau)\tilde{k}_T(\tau, s)$  are holomorphic on  $\mathbb{H}$ , which is simply connected; by Cauchy's theorem each integral is unchanged under any deformation fixing its endpoints, hence depends on those endpoints alone. This is path-independence.

For independence of  $\tau_0$  we differentiate the two integrals and watch them cancel. The *T*-segment runs from  $T^{-1}\tau_0 = \tau_0 - 1$  to  $\tau_0$  — two endpoints a unit apart, moving together with  $\tau_0$  — so by Leibniz's rule and the *T*-periodicity  $f(\tau_0 - 1) = f(\tau_0)$ ,

$$\partial_{\tau_0} \int_{\gamma^T} f \tilde{k}_T = f(\tau_0) \tilde{k}_T(\tau_0, s) - f(\tau_0 - 1) \tilde{k}_T(\tau_0 - 1, s) = f(\tau_0) \delta_T \tilde{k}_T(\tau_0 - 1, s), \quad (5)$$

where  $\delta_T \tilde{k}_T(\tau, s) := \tilde{k}_T(\tau + 1, s) - \tilde{k}_T(\tau, s)$ . This is the whole role of  $\delta_T$ : the unit gap between the endpoints turns the  $\partial_{\tau_0}$  boundary term into a unit shift of the kernel. That shift is elementary — the kernel was built for it — since  $\zeta(\nu, x + 1) - \zeta(\nu, x) = -x^{-\nu}$  gives, on both terms at  $x = \tau + 1$ ,

$$\delta_T \tilde{k}_T(\tau, s) = -(\tau + 1)^{s-1} + e^{i\pi(s-1)}(\tau + 1)^{k-1-s} \quad (6)$$

$$\implies \partial_{\tau_0} \int_{\gamma^T} f \tilde{k}_T = f(\tau_0) [-\tau_0^{s-1} + e^{i\pi(s-1)}\tau_0^{k-1-s}]. \quad (7)$$

The *S*-segment runs  $-1/\tau_0 \rightarrow \tau_0$ ; its lower endpoint moves with  $\partial_{\tau_0}(-1/\tau_0) = \tau_0^{-2}$ , so by Leibniz, the modularity  $f(-1/\tau_0) = \tau_0^k f(\tau_0)$ , and the phase  $(-1/\tau_0)^{s-1} = e^{i\pi(s-1)}\tau_0^{1-s}$ ,

$$\partial_{\tau_0} \int_{\gamma^S} f \tau^{s-1} = f(\tau_0)\tau_0^{s-1} - f(-1/\tau_0)(-1/\tau_0)^{s-1}\tau_0^{-2} = f(\tau_0)[\tau_0^{s-1} - e^{i\pi(s-1)}\tau_0^{k-1-s}]. \quad (8)$$

The two brackets are negatives of each other, so  $\partial_{\tau_0} L^* = e^{-\pi i s/2}(\partial_{\tau_0} \int_{\gamma^S} + \partial_{\tau_0} \int_{\gamma^T}) = 0$ .  $\square$

**Remark 1.7** (Classical *L*-function contour). Let  $f \in S_k$ . As  $\tau_0 \rightarrow i\infty$ , the *S*-segment maps onto the standard *L*-function contour,  $[0, i\infty]$ . Simultaneously, as  $f(it) \sim e^{-2\pi t}$  as  $t \rightarrow \infty$ , the *T*-segment vanishes exponentially. Thus, in this limit,  $L^*(f, s)$  exactly maps onto the classical *L*-function integral [15]:

$$\lim_{\tau_0 \rightarrow i\infty} L^*(f, s) = \int_0^\infty t^{s-1} f(it) dt. \quad (9)$$

## 2 $L$ -functions for weak modular forms

**Proposition 2.1** (Hurwitz-Gamma). *For  $n \neq 0$  and  $\nu \in \mathbb{C}$ ,*

$$\int_i^{i+1} e^{2\pi i n u} \zeta(\nu, u) du = e^{i\pi(1-\nu)/2} (2\pi n)^{\nu-1} \Gamma(1-\nu, 2\pi n). \quad (10)$$

*Proof.* Write  $\zeta(\nu, u) = \sum_{m \geq 0} (m+u)^{-\nu}$  and shift  $u \mapsto u+m$  in the  $m$ -th term; as  $e^{2\pi i n(u+m)} = e^{2\pi i n u}$  for integer  $m$ , the unit segments stack into one ray,

$$\int_i^{i+1} e^{2\pi i n u} \zeta(\nu, u) du = \sum_{m \geq 0} \int_{i+m}^{i+1+m} e^{2\pi i n v} v^{-\nu} dv = \int_i^{i+\infty} e^{2\pi i n v} v^{-\nu} dv. \quad (11)$$

For  $n > 0$  the integrand is holomorphic in the first quadrant and decays as  $\text{im } v \rightarrow \infty$  (since  $|e^{2\pi i n v}| = e^{-2\pi n \text{im } v}$ ), so the horizontal ray rotates to the vertical  $[i, i\infty)$ . With  $v = it$  ( $t \geq 1$ ; then  $dv = i dt$ ,  $e^{2\pi i n v} = e^{-2\pi n t}$ ,  $v^{-\nu} = e^{-\pi i \nu/2} t^{-\nu}$ ) and  $w = 2\pi n t$ ,

$$\int_i^{i+\infty} e^{2\pi i n v} v^{-\nu} dv = e^{i\pi(1-\nu)/2} \int_1^\infty e^{-2\pi n t} t^{-\nu} dt = e^{i\pi(1-\nu)/2} (2\pi n)^{\nu-1} \Gamma(1-\nu, 2\pi n). \quad (12)$$

For  $n < 0$  the integrand instead grows as  $\text{im } v \rightarrow \infty$ , so the closing arc survives and the rotation is illegal; but both sides are analytic in  $n$  and agree for  $n > 0$ , so the identity continues to  $n < 0$ , with  $\Gamma(1-\nu, 2\pi n)$  on its principal branch. At  $n = 0$  the prefactor  $(2\pi n)^{\nu-1}$  is singular — that mode, the constant Fourier coefficient, is handled separately.  $\square$

**Theorem 2.2** ( $L$ -function for weakly holomorphic forms). *For  $f \in M_k^!$ , we may set  $\tau_0 = i$ , and prove*

$$L^*(f, s) = -c_f(0) \left( \frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left( \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right). \quad (13)$$

*This  $L$ -integral satisfies the functional equation  $L^*(f, s) = i^k L^*(f, k-s)$ .*

*Proof.* By path independence and base-point independence from Proposition 1.6, we may freely set  $\tau_0 = i$ . Here, the  $S$ -segment degenerates ( $-1/i = i$ ), and the  $L$ -integral simplifies dramatically:

$$L^*(f, s) = e^{-\pi i s/2} \int_{i-1}^i f \tilde{k}_T d\tau. \quad (14)$$

We then expand  $f$  in terms of its  $q$ -series,  $\sum_n c_f(n) e^{2\pi i n \tau}$ , and substitute  $u = \tau + 1$ , and note that  $e^{2\pi i n(u-1)} = e^{2\pi i n u}$ . Here, each  $n \neq 0$  mode is handled by Proposition 2.1. The first kernel term with  $\nu = 1-s \leftrightarrow 1-\nu = s$  gives

$$e^{-\pi i s/2} e^{\pi i s/2} (2\pi n)^{-s} \Gamma(s, 2\pi n) = \Gamma(s, 2\pi n) / (2\pi n)^s, \quad (15)$$

and the second kernel term with  $\nu = s-k+1 \leftrightarrow 1-\nu = k-s$  gives

$$-e^{-\pi i s/2} e^{i\pi(s-1)} e^{\pi i(k-s)/2} (2\pi n)^{s-k} \Gamma(k-s, 2\pi n) = i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}}. \quad (16)$$

So each  $n \neq 0$  mode contributes  $\Gamma(s, 2\pi n)/(2\pi n)^s + i^k \Gamma(k-s, 2\pi n)/(2\pi n)^{k-s}$ . Crucially, the  $n = 0$  mode contributes

$$c_f(0) e^{-\pi i s/2} \int_{i-1}^i \tilde{k}_T d\tau = -c_f(0) \left( \frac{1}{s} + \frac{i^k}{k-s} \right). \quad (17)$$

Combining these reproduces Eq. (13). Finally, the functional equation,  $L^*(f, s) = i^k L^*(f, k-s)$ , follows by direct inspection.  $\square$

## 2.1 Poles at the cusp vs the classical Mellin contour

## 2.2 Classical Mellin recovery and canonical uniqueness.

For  $f \in S_k$  a holomorphic cusp form, taking the cusp-limit  $\tau_0 \rightarrow i\infty$  in Theorem 2.2 collapses the  $S$ -segment onto the classical Mellin contour  $[0, i\infty]$  while the  $T$ -segment vanishes exponentially (since  $f(\tau) = O(e^{-2\pi \operatorname{im} \tau})$  on the segment as  $\operatorname{im} \tau \rightarrow \infty$ ):

$$\lim_{\tau_0 \rightarrow i\infty} L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = \int_0^\infty t^{s-1} f(it) dt, \quad (18)$$

the classical completed  $L$ -integral of  $f$ . For  $f \in M_k$  with  $c_f(0) \neq 0$  the same limit holds after a standard constant-term subtraction (see [4, Lemma 2.1] for the explicit polynomial correction).

For  $f \in M_k^!$  with cusp poles the cusp-limit is genuinely pathological:  $f(\tau) \rightarrow \infty$  as  $\operatorname{im} \tau \rightarrow \infty$ , so the  $S$ -segment integral diverges along the imaginary axis, and the  $T$ -segment integral  $\int_{\tau_0-1}^{\tau_0} f(\tau) \tilde{k}_T(\tau, s) d\tau$  acquires a matching divergence of the opposite sign as  $\operatorname{im} \tau_0 \rightarrow \infty$  — made precise in Lemma ??, where the  $T$ -segment is shown to equal an explicit incomplete-gamma sum whose polar mode  $q^{-M}$  contributes a term growing like  $|c_f(-M)| e^{2\pi M \operatorname{im}(\tau_0)}$ . The two divergences cancel at every finite  $\tau_0$ , so  $L^*(f, s)$  is finite and well defined throughout  $\mathbb{H}$ , but no naive Mellin contour limit exists. By throwing-out  $\gamma_{\tau_0}^T$  at the outset, one does violence to the result; the  $\tau_0 \rightarrow i\infty$  limit of the  $S$ -contour alone diverges, necessitating regularizations of the type introduced in [1]. The construction of [1] recovers  $L^*(f, s)$  on  $M_k^!$  by a non-holomorphic regulator at the cusps; Theorem 2.2 shows that this regulator and the finite-contour functional in Definition 1.3 compute the same object, term by term in the Fourier expansion. In particular,  $L^*(f, s)$  is independent of any choice of regularization: it is the value of the finite-contour functional at any admissible  $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ , and the BFK construction is one closed-form evaluation of it (at  $\tau_0 = i$ ).

## 2.3 Period polynomials, periods and quasi-periods

Because  $L^*(f, s)$  is canonical for every  $f \in M_k^!$ , the period polynomial needs no separate construction — it is the generating polynomial of the critical values.

**Definition 2.3** (Period polynomial). *For  $f \in S_k^!$  and  $n = k - 2$ , the period polynomial is*

$$r_f(X, Y) := (2\pi i)^{n+1} \sum_{\ell=0}^n (-1)^\ell \binom{n}{\ell} i^{\ell+1} L^*(f, \ell+1) X^{n-\ell} Y^\ell. \quad (19)$$

**Proposition 2.4** (Period polynomial as a contour integral). *With  $(X - \tau Y)^n$  in place of the  $L$ -weight  $\tau^{s-1}$ , and  $\mathbf{K}_T(\tau; X, Y) := \sum_{\ell=0}^n \binom{n}{\ell} (-1)^\ell \tilde{k}_T(\tau, \ell+1) X^{n-\ell} Y^\ell$  in place of  $\tilde{k}_T$ , the same two segments assemble  $r_f$  directly:*

$$r_f = (2\pi i)^{n+1} \left[ \int_{\gamma_S} f(\tau) (X - \tau Y)^n d\tau + \int_{\gamma_T} f(\tau) \mathbf{K}_T(\tau; X, Y) d\tau \right]. \quad (20)$$

*Proof.* By Definition 1.3, the  $X^{n-\ell} Y^\ell$  coefficient of the bracket is

$$r_f|_{X^{n-\ell} Y^\ell} = (-1)^\ell \binom{n}{\ell} \left[ \int_{\gamma_S} f \tau^\ell + \int_{\gamma_T} f \tilde{k}_T(\tau, \ell+1) \right] = (-1)^\ell \binom{n}{\ell} i^{\ell+1} L^*(f, \ell+1). \quad (21)$$

This matches definition 2.3 term by term, and proves the claim.  $\square$

**Quasi-periods and their determinant:** As an aside, the Hurwitz T-kernel in this note was originally uncovered in order to reproduce the manifestly finite computations of the quasi-periods for  $\widehat{\Delta} = 1/q + O(q^2) \in S_{12}^!$  originally given in [2]. In this vein, it is interesting to show collect quasi-periods computable from our L-integrals/period-polynomials for  $f \in S_k^!$  (when  $\dim(S_k) = 1$ ).

To this effect, we recall the notation of [6]. Decomposing the modular weight  $k = 12\lambda + k'$  where  $k' \in \{0, 4, 6, 8, 10, 14\}$ , we see  $\lambda = \dim(S_k)$  when  $\lambda \geq 0$ . For any  $\lambda$ , we may denote  $f_{k,m} := 1/q^m + O(q^{\lambda+1}) \in M_k^!$ . When  $\dim(S_k) = 1$ , we have  $f_{k,m} = 1/q^m + O(q^2)$ . Here, define  $\Delta_k := f_{k,-1} = q + O(q^2) \in S_k$  and  $\widehat{\Delta}_k := f_{k,1} = 1/q + O(q^2) \in S_k^!$ . Further, define

$$\text{pd}_k := \begin{pmatrix} \omega^+(\Delta_k) & \omega^-(\Delta_k) \\ \eta^+(\widehat{\Delta}_k) & \eta^-(\widehat{\Delta}_k) \end{pmatrix} \quad (22)$$

That  $\det(\text{pd}_k)$  satisfies a Legendre-type relation,  $\det(\text{pd}_k) \in \mathbb{Q}^\times \cdot (2\pi i)^{k-1}$ , is the de Rham theory of Brown–Hain [3, Prop. 5.6]. Brown [2, §8.3] evaluates the constant at  $k = 12$ , finding  $\det(\text{pd}_k) = -(k-2)!(2\pi i)^{k-1}$  — his period matrix orders the quasi-periods first, giving  $+(k-2)!$ , where ours orders the periods first — and remarks that he could locate no reference for the value. We reproduce  $k = 12$  and give the analogous rational multiples for  $k \in \{16, 18, 20, 22, 26\}$  in Table 1, below:

| $k$ | $\omega^+(f_k)$ | $\omega^-(f_k)/i$ | $\eta^+(\widehat{\Delta}_k)$ | $\eta^-(\widehat{\Delta}_k)/i$ | $\frac{-i^k \det(\text{pd}_k)}{(2\pi)^{k-1}(k-2)!}$ |
|-----|-----------------|-------------------|------------------------------|--------------------------------|-----------------------------------------------------|
| 12  | −6.89168e7      | −5.58502e6        | 1.27202e11                   | 1.02767e10                     | 1                                                   |
| 16  | −8.81481e11     | −1.54063e10       | 4.14120e13                   | 6.83541e11                     | 13/30                                               |
| 18  | −5.07833e13     | −5.43305e12       | −2.19200e14                  | −3.36339e13                    | 2/3                                                 |
| 20  | −4.22882e16     | −3.73170e14       | 6.38640e16                   | −1.89823e14                    | 17/5                                                |
| 22  | −1.49974e18     | −4.24489e16       | 1.43795e18                   | −4.41050e16                    | 19/21                                               |
| 26  | −1.18673e22     | −3.60961e21       | 1.18653e22                   | −3.61016e21                    | 23/15                                               |

Table 1: Periods  $\omega^\pm(\Delta_k)$  and quasi-periods  $\eta^\pm(\widehat{\Delta}_k)$ , and the rational number  $(-1)^{1+k/2} \det / ((2\pi i)^{k-1}(k-2)!)$  with  $\det(\text{pd}_k) = \omega^+ \eta^- - \omega^- \eta^+$ . At  $k = 12$  the value matches Brown [2]; the entries for  $k \geq 16$  appear to be new. These ratios are not intrinsic to  $f$ : they depend on how the rational period-polynomial basis  $P^\pm$  is normalized (we use a primitive-integer normalization; Brown’s normalization gives  $-(k-2)!$  at  $k = 12$ ). We have not found a closed form for them. Full 50-digit values in [5].

### 3 Meromorphic modular forms

We find it convenient to define the following function,

$$G_{s,k}(n) := \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}}. \quad (23)$$

**Proposition 3.1** (Basepoint and path independence within a homotopy class). *For  $f \in F_k$ , the value  $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  is independent of  $\tau_0$  and of the path so long as both vary within a single homotopy class of admissible contours in  $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ . For  $f \in M_k^!$  there is a single class, recovering Proposition 1.6; across different classes the value changes by the residues the contour sweeps, as quantified in Corollary 3.9.*

*Proof.* The  $\tau_0$ -derivative of Proposition 1.6 uses only the  $T$ -periodicity  $f(\tau_0 - 1) = f(\tau_0)$  and the modularity  $f(-1/\tau_0) = \tau_0^k f(\tau_0)$ , both valid for every  $f \in F_k$ , and is evaluated at the moving endpoints alone. While  $\tau_0$  stays in one chamber of  $\mathbb{H} \setminus f^{-1}(\infty)$  — where the segments meet no pole — the boundary terms cancel exactly as there, so  $\partial_{\tau_0} L^* = 0$ ; the interior poles lie off the contour

and never enter. Within a fixed class the integrands  $f\tau^{s-1}$  and  $f\tilde{k}_T$  are holomorphic along the contours, so any two are joined by a deformation avoiding  $\Gamma \cdot f^{-1}(\infty)$ , and Cauchy's theorem equates their integrals.  $\square$

**Definition 3.2** (Polylog projection). *Let  $f \in F_k$  have interior poles at the points  $\tau_p$  of a set  $\mathcal{P}_f$  of  $\Gamma$ -inequivalent representatives, and write  $\text{Li}_s(z) = \sum_{n>0} z^n n^{-s}$ . Since  $\text{Li}_{1-m}(e^{2\pi i(\tau-\tau_p)})$  has a pole of order  $m$  at  $\tau_p$ , the principal part of  $f$  there is matched by a unique polylog projection*

$$\hat{P}_{\tau_p} f(\tau) := \sum_{m>0} r_{f,\tau_p}^*(m) \text{Li}_{1-m}(e^{2\pi i(\tau-\tau_p)}), \quad (24)$$

*its coefficients  $r_{f,\tau_p}^*(m)$  being the principal-part data of  $f$  at  $\tau_p$ . The projection  $\hat{P} := \sum_{\text{Im } \tau_p \geq 1} \hat{P}_{\tau_p}$  removes the principal part of every pole on or above the line  $\text{Im } \tau = 1$ .*

**Definition 3.3** (Regularized cusp coefficients). *With  $\hat{P}$  as in Definition 3.2,*

$$c_{Rf}(n) := [(I - \hat{P})f]_{q^n}. \quad (25)$$

**Definition 3.4** (The space  $F(\infty)$ ).  *$F(\infty)$  is the space of meromorphic  $g$  on  $\mathbb{H}$  with  $g(\tau) = g(\tau+1)$ , finitely many poles in any compact subset of the strip  $\{\text{Im } \tau > 0, |\text{Re } \tau| \leq \frac{1}{2}\}$ , and  $g(\tau) = O(e^{C \text{Im } \tau})$  as  $\text{Im } \tau \rightarrow \infty$  for some  $C$ . Each such  $g$  has a Fourier expansion  $g = \sum_n c_g(n) q^n$ .*

Then  $F_k \subset F(\infty)$ , and  $(I - \hat{P})f \in F(\infty)$  — the polylog tails are  $T$ -periodic but not modular, so the projection leaves  $F_k$ .

**Lemma 3.5** ( $q$ -series–integral commutation). *Let  $g \in F(\infty)$  have no pole on or above the line  $\text{Im } \tau = 1$  (e.g.  $g = (I - \hat{P})f$ ). Then  $g = \sum_n c_g(n) q^n$  converges along the  $T$ -segment at  $\tau_0 = i$  and integrates term by term:*

$$\int_{\gamma_i^T} g(\tau) \tilde{k}_T(\tau, s) d\tau = \sum_n c_g(n) \int_{i-1}^i e^{2\pi i n \tau} \tilde{k}_T(\tau, s) d\tau. \quad (26)$$

*A pole at  $\text{Im } \tau_p \geq 1$  defeats this — there the coefficients grow at least like  $n^{m-1} e^{2\pi n \text{Im } \tau_p}$  [4, Lem. 2.6].*

*Proof.* With no pole on or above  $\text{Im } \tau = 1$ , the finitely many poles of  $g$  in the strip have heights  $< 1$ ; let  $h < 1$  be the largest. The Fourier series  $g = \sum_n c_g(n) q^n$  converges absolutely and uniformly on compact subsets of  $\{\text{Im } \tau > h\}$ , in particular on the  $T$ -segment  $[i-1, i]$ ; with  $\tilde{k}_T(\tau, s)$  bounded there, the sum passes through the integral, which is the displayed identity. Conversely a pole at height  $\text{Im } \tau_p \geq 1$  forces  $|c_g(n)| \gtrsim n^{m-1} e^{2\pi n \text{Im } \tau_p}$  [4, Lem. 2.6], so  $|c_g(n) q^n| \gtrsim n^{m-1} e^{2\pi n (\text{Im } \tau_p - 1)} \not\rightarrow 0$  on  $[i-1, i]$  and the series diverges.  $\square$

**Definition 3.6** (Polylog integral). *For  $N \geq 0$  and  $z \in \mathbb{H}$ , the polylog integral is the  $L$ -integral of a single tail  $\text{Li}_{-N}(e^{2\pi i(\tau-z)})$  along the  $T$ -segment at  $\tau_0 = i$  (where  $\gamma^S$  degenerates, since  $-1/i = i$ ):*

$$I_N(s, z) := e^{-\pi i s/2} \int_{i-1}^i \text{Li}_{-N}(e^{2\pi i(\tau-z)}) \tilde{k}_T(\tau, s) d\tau. \quad (27)$$

**Lemma 3.7** (Specialized polylog integral). *Recall  $G_{s,k}(n)$  from Eq. (23). For the projected poles  $\text{Im } z > 1$ , where  $|e^{2\pi i(\tau-z)}| \geq 1$  on the contour, polylog inversion  $\text{Li}_{-N}(w) = (-1)^{N+1} \text{Li}_{-N}(1/w)$  resums the integral into the convergent series*

$$I_N(s, z) = \delta_{N,0} \left( \frac{1}{s} + \frac{i^k}{k-s} \right) + (-1)^{N+1} \sum_{n \geq 1} n^N e^{2\pi i n z} G_{s,k}(-n). \quad (28)$$

*This sum converges exponentially quickly, with base  $e^{-2\pi n(y-1)}$ , where  $y = \text{Im}(z)$ .*

*Proof.* For  $\text{Im } z > 1$ ,  $|e^{2\pi i(\tau-z)}| \geq 1$  on  $[i-1, i]$ , so the series of  $\text{Li}_{-N}(e^{2\pi i(\tau-z)})$  diverges; the inversion rewrites it through  $\text{Li}_{-N}(e^{2\pi i(z-\tau)}) = \sum_{n \geq 1} n^N e^{2\pi i n(z-\tau)}$ , convergent since  $|e^{2\pi i(z-\tau)}| \leq 1$ . Integrating term by term against  $\tilde{k}_T$ , each  $e^{-\pi i s/2} \int_{i-1}^i e^{-2\pi i n \tau} \tilde{k}_T d\tau$  is the building block  $G_{s,k}(n)$  of Theorem 2.2 at mode  $-n$ , giving the displayed sum; for  $N = 0$  the leftover  $-1$  in  $\text{Li}_0(w) = -1 - \text{Li}_0(1/w)$  adds  $-e^{-\pi i s/2} \int_{i-1}^i \tilde{k}_T = \frac{1}{s} + \frac{i^k}{k-s}$ .

Exponential convergence  $\sim e^{-2\pi n(y-1)}$  for  $y = \text{Im}(z) > 1$  follows from combining the asymptotic scaling  $\Gamma(s, -2\pi n) \sim (2\pi n)^{s-1} e^{2\pi n}$  and the exact result  $|e^{2\pi i n z}| = e^{-2\pi n y}$ .  $\square$

**Theorem 3.8** (*L*-function of a meromorphic modular form). *For  $f \in F_k$  with no pole on the edge  $\text{Im } \tau = 1$ , at  $\tau_0 = i$ ,*

$$L^*(f, s) = -c_{Rf}(0) \left( \frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_{Rf}(n) G_{s,k}(n) + \sum_{\text{Im } \tau_p > 1} \sum_{m > 0} r_{f, \tau_p}^*(m) I_{m-1}(s, \tau_p), \quad (29)$$

where, again  $G_{s,k}(n)$  is the sum of incomplete gammas defined in Eq. (23); the regularized cusp sum (Definition 3.3, Theorem 2.2) plus one polylog integral (Lemma 3.7) per interior pole above the edge. A pole on the edge  $\text{Im } \tau = 1$  — the elliptic point  $\tau = i$  included — instead sits on the contour: there  $L^*$  is the principal value, and sweeping the contour across the pole changes it by the residue of Corollary 3.9.

*Proof.* At  $\tau_0 = i$  the  $S$ -segment degenerates, so  $L^*(f, s) = e^{-\pi i s/2} \int_{i-1}^i f \tilde{k}_T d\tau$ . Split  $f = (I - \hat{P})f + \hat{P}f$ . The regularized part has no pole on or above  $\text{Im } \tau = 1$  (Definition 3.2), so by Lemma 3.5 its Fourier series integrates term by term; mode by mode this is Theorem 2.2, giving  $-c_{Rf}(0) \left( \frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_{Rf}(n) G_{s,k}(n)$ . The projected part  $\hat{P}f = \sum_{\tau_p} \sum_{m > 0} r_{f, \tau_p}^*(m) \text{Li}_{1-m}(e^{2\pi i(\tau-\tau_p)})$  integrates to  $\sum_{\text{Im } \tau_p > 1} \sum_{m > 0} r_{f, \tau_p}^*(m) I_{m-1}(s, \tau_p)$  by Lemma 3.7. Their sum is the stated formula.  $\square$

**Corollary 3.9** (Wall-crossing). *By Proposition 3.1,  $L^*(f, s)$  is fixed within each homotopy class. Crossing a wall, where the contour sweeps past an interior pole  $\tau_p$ , changes it by*

$$\Delta L^*(f, s) = 2\pi i e^{-\pi i s/2} \text{Res}_{\tau=\tau_p} [f(\tau) \tilde{k}_T(\tau, s)] \quad (30)$$

when the  $T$ -segment crosses  $\tau_p$  (and by the same with  $\tau^{s-1}$  in place of  $\tilde{k}_T$  when the  $S$ -segment does). Thus  $L^*$  is fixed by the contour up to the residues it crosses.

*DAM: I do not think that this result is correct/general enough: The  $S$ -contour or the  $T$ -contour could enclose poles, and the residues would be different. Note that, unlike for the case of  $f \in M_k^1$ , the fact that the  $S$ -contour has end-points that coincide, the contribution can be non-zero. Indeed, that is exactly the case for enclosing a pole in a closed contour!*

*Proof.* The within-class constancy is Proposition 3.1. Crossing a wall, the contour sweeps once past  $\tau_p$ ; the residue theorem then adds  $2\pi i e^{-\pi i s/2} \text{Res}_{\tau=\tau_p} [f \tilde{k}_T]$  when the  $T$ -segment crosses it, or the same with  $\tau^{s-1}$  when the  $S$ -segment does.  $\square$

## 4 To do list...

1. DAM: Need to make sure each bibitem is relevant for this doc!!!
2. Reinstate the "why did BFK have to regularize? BECAUSE THEY DIDN'T USE THE CORRECT CONTOUR AND THREW OUT THE DIVERGENT T-PIECE WITHOUT EVEN KNOWING!"-section

3. Should find a better definition of the homotopy class. Like,  $L^*(f, s; X_S, X_T)$  where  $X_S = \{$  the set of all windings of all poles of  $f$  enclosed by the S-contour, and  $X_T$  does the same for the T-contour.
4. Make spellings American, not British
5. other stuff...

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